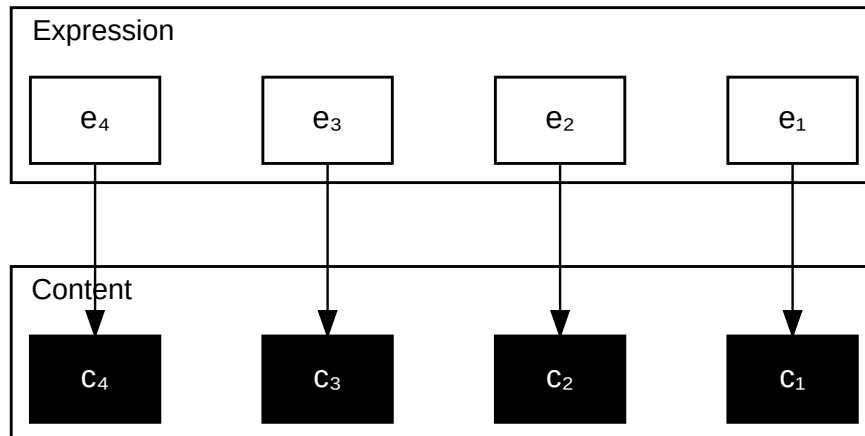


01

The Code

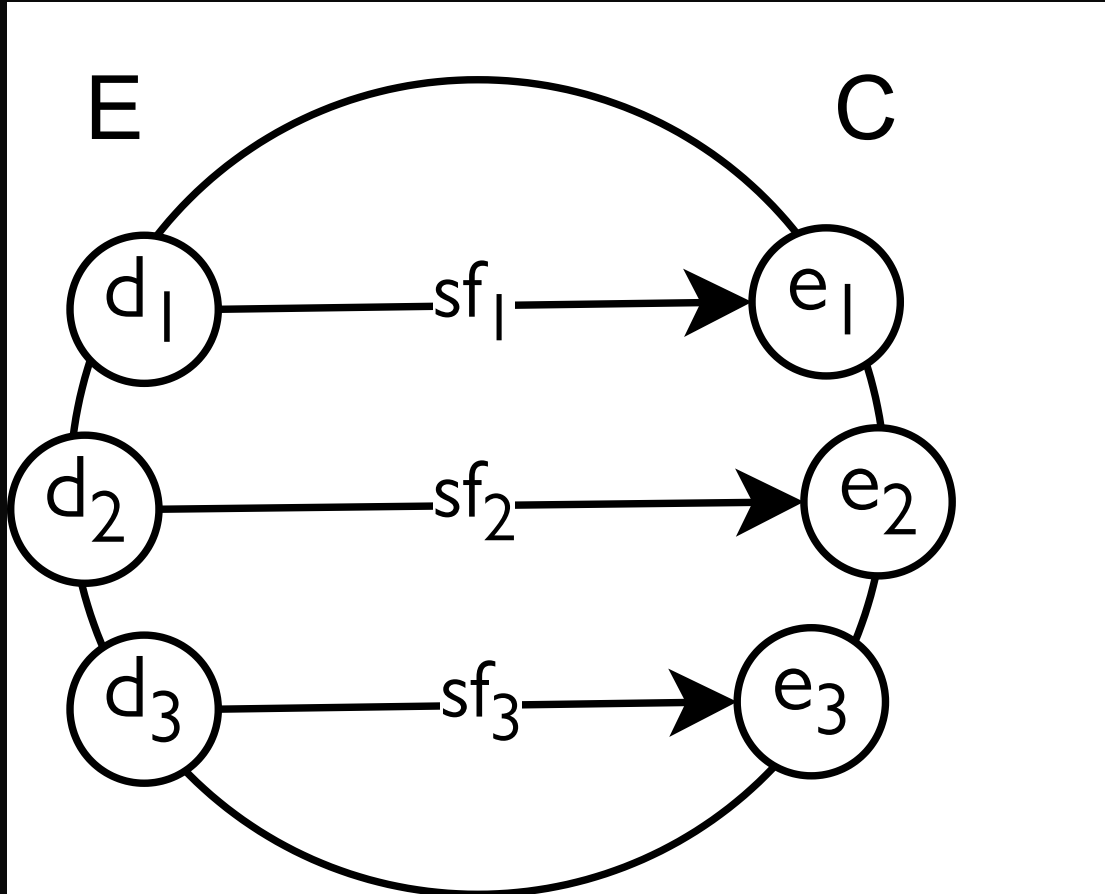
Code



ECO 1975

The code as bijection

Each element of Expression is put in relation with one element of Content. Adequate for ant chemotaxis, highway codes, formal grammars.



HOLLAND · ADAPTIVE AGENTS

Embodying the code

- Code = rules governing **detector / effector** relations (**sign functions**)
- Content resolves into action — **sign production**
- Detectors and effectors are **types**: matching and instantiation

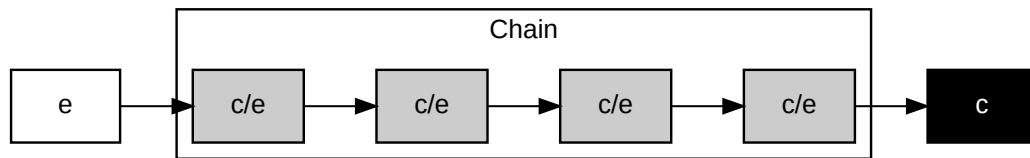
HJELMSLEV · TWO READINGS

Expression / Content: two interpretations in tension.

- **Substantial** — phonology vs. semantics; the Saussurean inheritance
- **Formal** — trigger vs. response; input type / output type
- The formal reading enables the law of total E/C reversibility

02

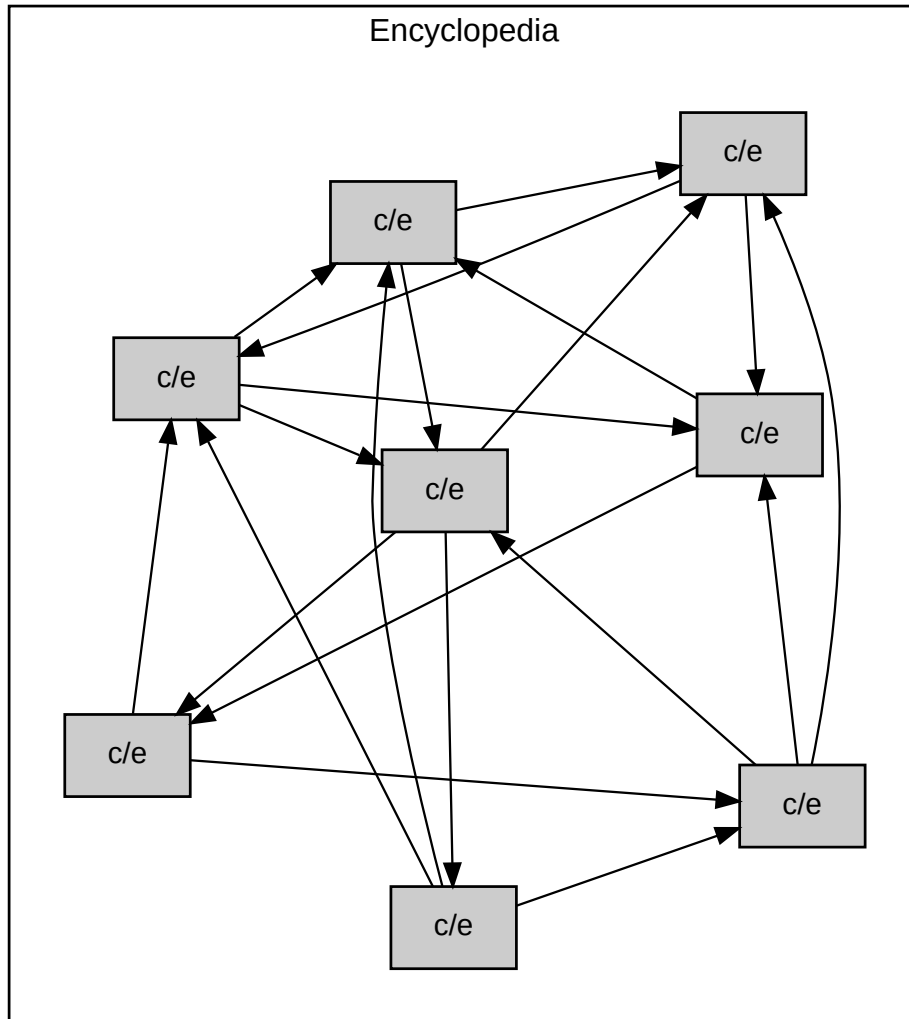
From Code to Encyclopedia



LINEAR TOPOLOGY

The chain

A content token output by one agent becomes an expression token for the next. Expression and content are *positions* assumed by forms of material substances.



ECO 1984 · MODEL Q

The encyclopedia

A strongly connected graph linking units in a non-univocal manner. The dictionary is a degenerate case — an encyclopedia with simplified topology.

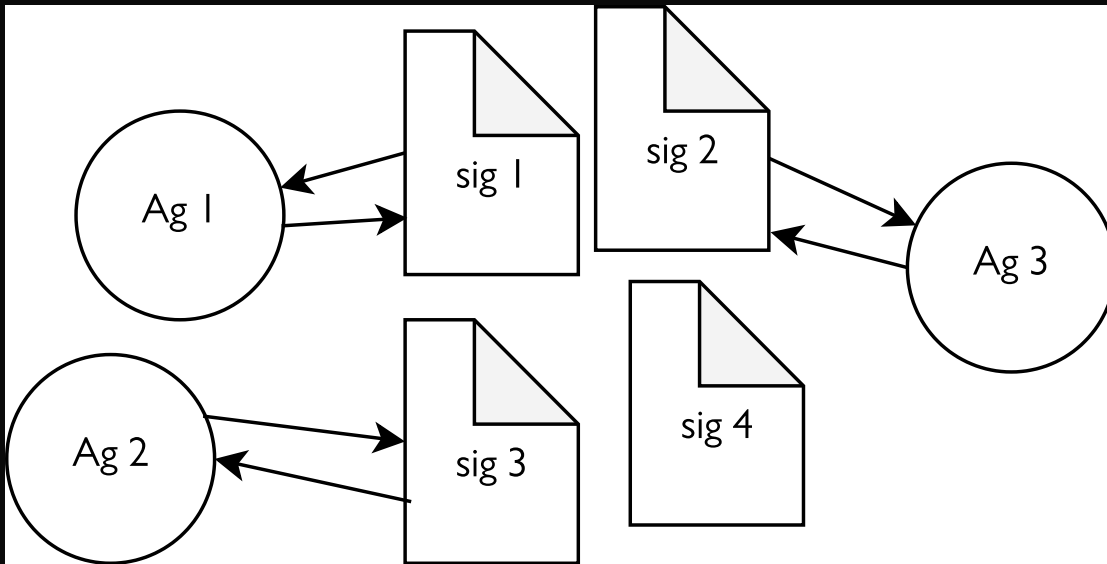
PERIMETRAL CONDITIONS · LIMINAL REALISM

Two problems at the boundary of semiosis:

- **How does it start?** — The prime mover must lie outside semiosis. The subject is thrown into a world.
- **How does it stop?** — Semiosis costs energy and time. Not "who speaks?" but *who dies?*

03

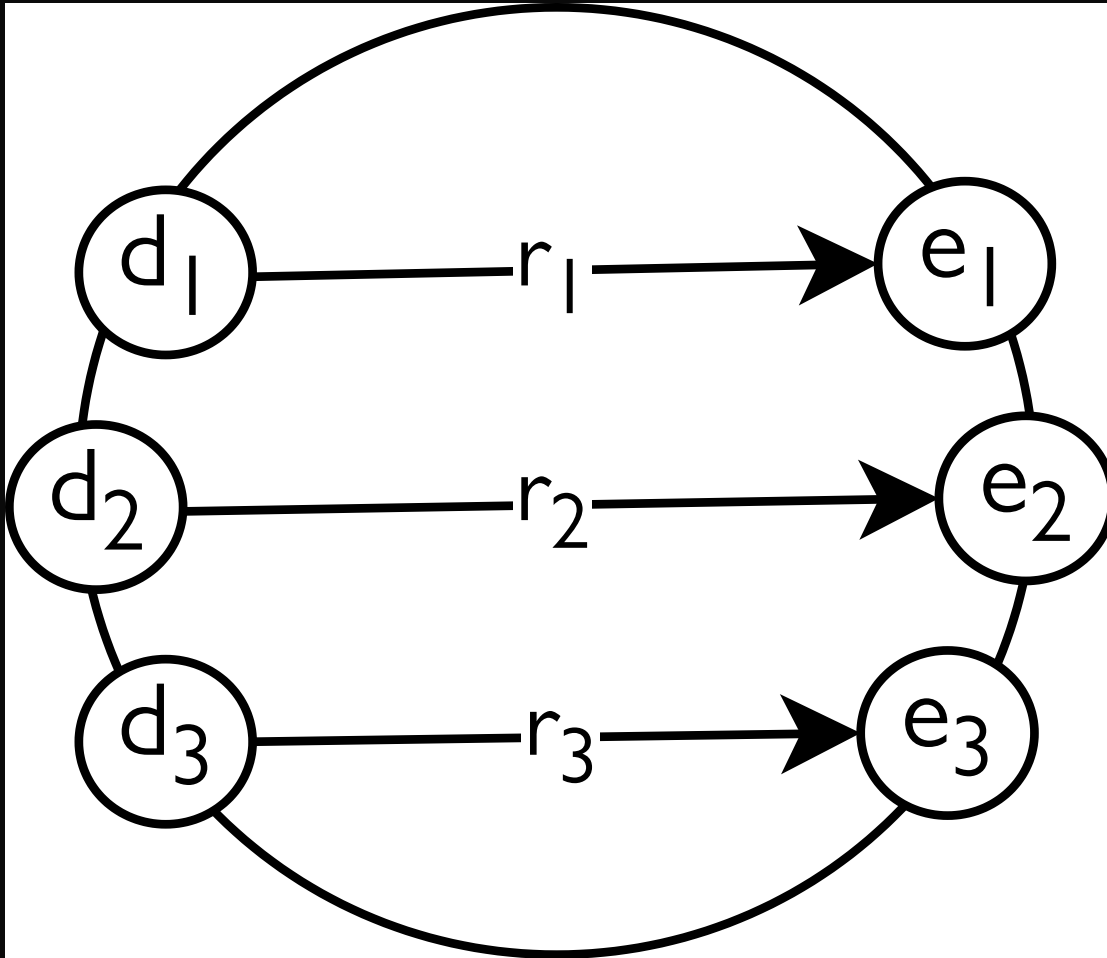
Signal Soup



THE MODEL

Environment & Agents

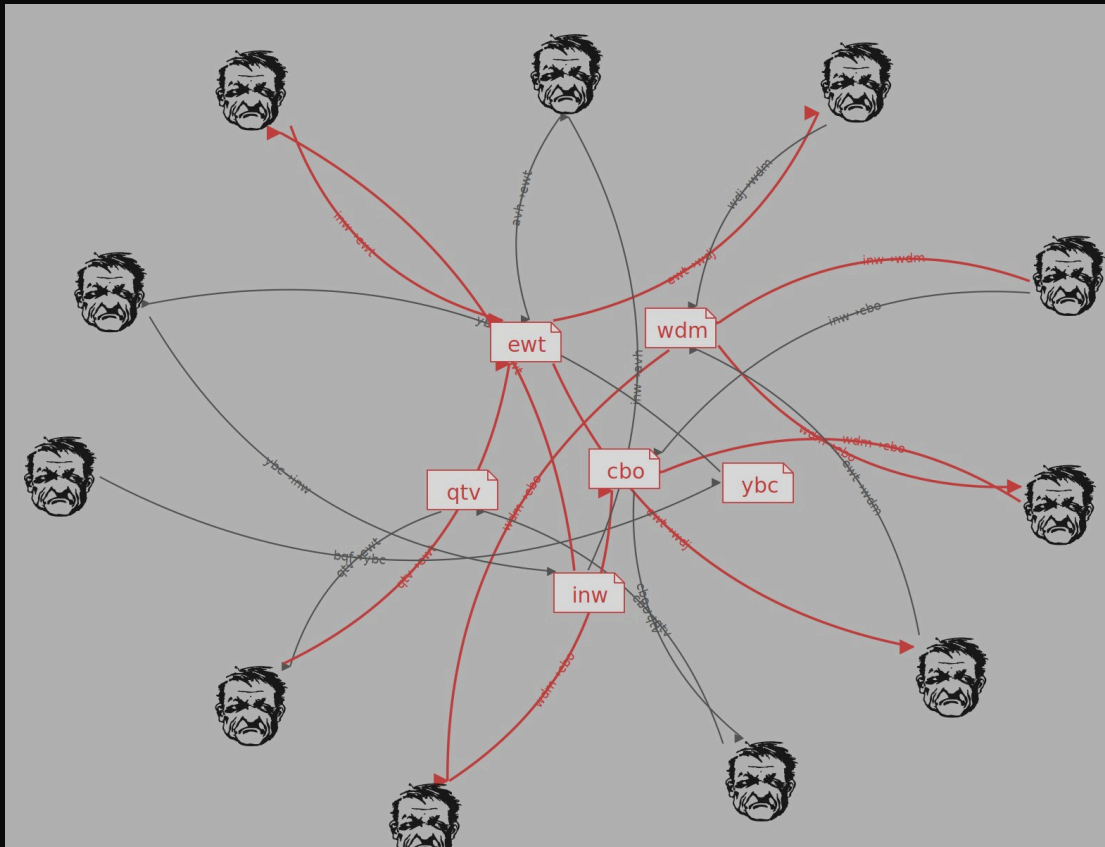
- Coordination through **stigmergy** — indirect, no direct feedback
- **Polylogical**, not dialogical — no dyadic pairs
- Signals: 3-letter strings → 17,576 possible types
- Each agent deals with a *signal soup* created by the other agents



AGENT ALGORITHM

Sign functions $sf : d \rightarrow e$

- **Match** — fire effector, replace signal in Environment, increment r
- **No match** — link two random signals into a new sf ($r = 1$)
- Memory cap $sf_{\max} = 10$ — drop lowest- r sf if exceeded



THE BAND

A population sharing an Environment

- Each Agent initialized with one random sf — $r = 0$, easily replaced
- Environment seeded with one signal per Agent — size = n
- **Asynchronous** — one randomly selected Agent per turn
- A small, tightly integrated communicative **circuit**

04

What Emerges

T = 0

T = 2,000

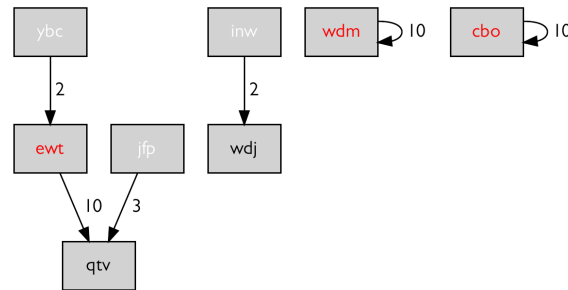
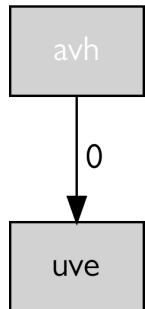


FIG. 2 · SINGLE AGENT TOPOLOGY

Complexification

- **Loops** — agent responds to its own output
- **Synonymy** — different detectors, same effector
- **Chains** — longer auto-response sequences

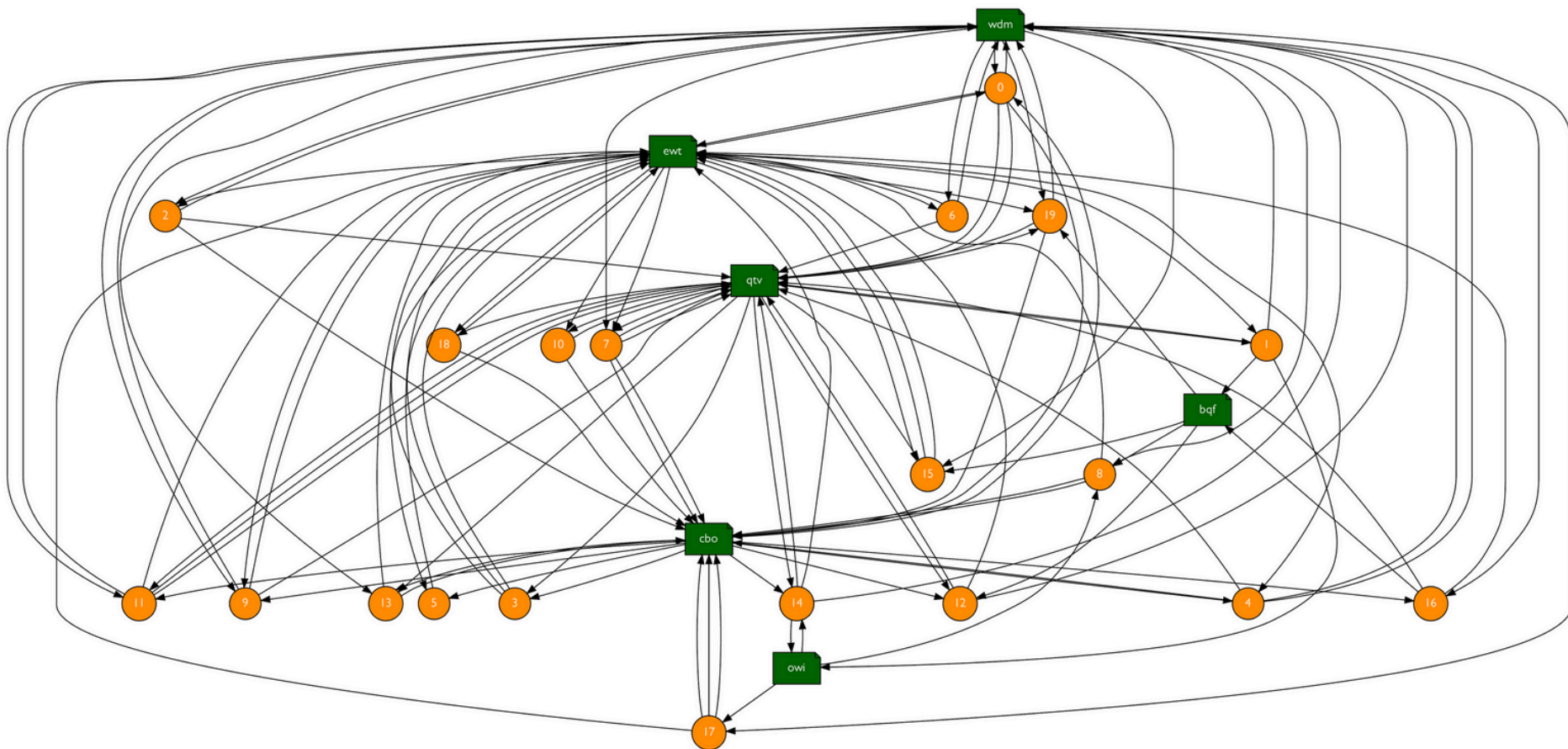
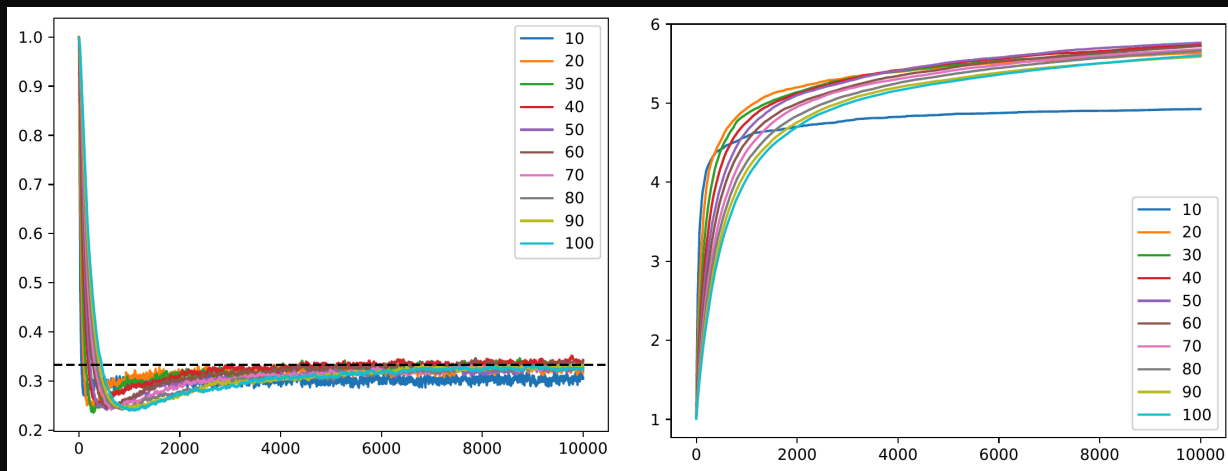


fig. 3 Agents' matching relations in Environment — 20-agent Band after 2,000 turns. Environment redundancy increases over time.

FIG. 4 · EMERGENT CONSTANTS



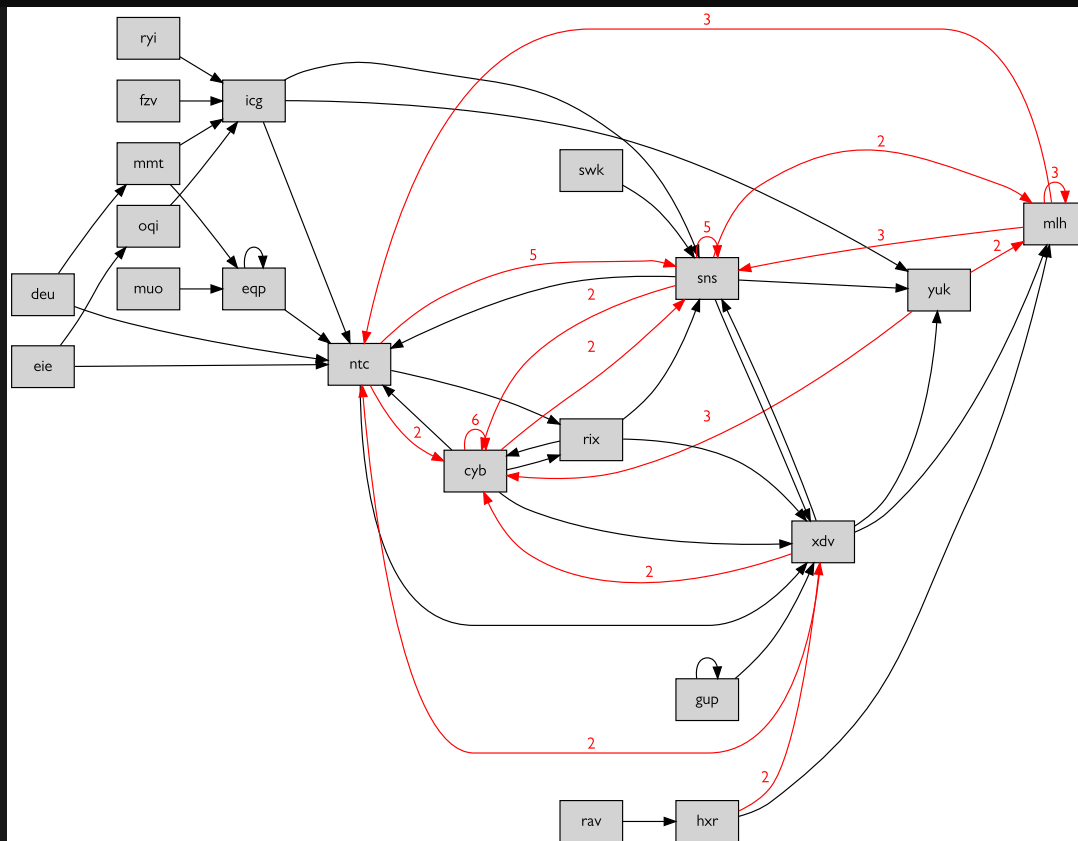
≈ 5.6

SIGN FUNCTIONS / AGENT

$1/3$

TYPE / TOKEN RATIO

Not programmed. Robust across all band sizes and parameter variations.



05

Conclusions

CONCLUSION 1

The encyclopedic topology is not prior to communication — it *emerges from* it.

- Each agent's graph crystallizes through interaction with Environment
- Environment is constituted by other agents' outputs
- The encyclopedia is endogenous to the communicative process

CONCLUSION 2

Sharedness operationalizes Eco's locality of the encyclopedia.

- Each agent holds a **local** encyclopedia — partial, idiosyncratic
- Sharedness = degree of overlap between local encyclopedias
- Full alignment (the dictionary) is the degenerate case: zero idiosyncrasy

GRAZIE · THANK YOU

Andrea Valle
Università di Torino

Inheriting Eco · Bologna 2026